

Title Slide

Predicting SpaceX Launch Success

A Data Science Capstone Project

Your Name | Date

Executive Summary

Goal: Predict the likelihood of SpaceX launch success.

Steps: Data collection, cleaning, EDA, visualization, predictive modeling.

Key Result: Achieved over 92% accuracy.

Introduction

SpaceX aims to make space travel affordable and reliable.

Predicting launch success helps mission planning.

Business Value: Cost savings, risk management.

Data Collection & Wrangling

Source: SpaceX API and Kaggle datasets.

Collected flight data, payloads, orbits, outcomes.

Handled missing values, standardized categories, created features.

EDA & Visual Analytics Methodology

Used Pandas, Matplotlib, Seaborn, Plotly.

Explored payload mass, orbit, success rates.

Developed interactive visualizations.

Predictive Analysis Methodology

Models: Logistic Regression, Decision Trees, Random Forest.

Metrics: Accuracy, Precision, Recall, ROC-AUC.

Performed hyperparameter tuning.

EDA Visualization Results

Bar charts: Launch site vs. success.

Scatter plots: Payload vs. success.

Heatmaps for correlations.

EDA with SQL Results

SQL queries:

- Launch site success counts
- Average payload by success

Results visualized with charts and tables.

Interactive Map with Folium

Folium map showing launch sites.

Markers with site details and success rates.

Included map screenshots.

Plotly Dash Dashboard

Interactive dashboard with filters:

- Payload mass
- Orbit type

Real-time success predictions.

Dashboard screenshots.

Predictive Analysis Results

Model accuracy:

- Logistic Regression: 89%

- Random Forest: 92%

Included confusion matrix & ROC curve.

Conclusion

High accuracy achieved.

Interactive tools provide insights.

Future: Add weather, timing features.

Creativity & Innovation

Thematic colors, animations.

Plotly & Folium visuals.

Suggested business insights.

References & GitHub Link

GitHub: [Your Link Here]

Sources: SpaceX API, Kaggle datasets.